

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
7	(a)		Use of $F = Ap$ and $A = \pi r^2$ or accept $A = 4\pi r^2$ (1) Correct answer = 3 173 [N] (1) [no ecf from use of $A = 4\pi r^2$]	1	1		2	2	
	(b)		Fewer collisions (1) ...because greater distances between molecules (or smaller density or more free space) (1)		2		2		
	(c)	(i)	Application of conservation of energy i.e. $E_k = \frac{Qq}{4\pi\epsilon_0 r}$ (1) Conversion of 4.7 MeV $\rightarrow$ J i.e. $4.7 \times 10^6 \times 1.6 \times 10^{-19} = 7.52 \times 10^{-13}$ J (1) Answer = 4.8×10^{-14} [m] (1)		3		3	3	
		(ii)	Smaller than atomic radius or inside plum pudding (1) So force / PE never great enough (for rebound) or scattering angle too large in experiment (1)			2	2		
	(d)		Use of conservation of energy to get speed or momentum e.g. $p^2 = 2mE_k$ etc. $v = 3.75 \times 10^7$ [m s ⁻¹] or $p = 3.41 \times 10^{-22}$ [N s] (1) Calculation of a wavelength using $\lambda = \frac{h}{p}$ (even if incorrect, 1.94×10^{-11} m is the correct value) (1) Comparison of the calculated wavelength with atomic separation (or 10^{-9} to 10^{-11} m) (1) Correct final conclusion and correct wavelength (1.94×10^{-11} m) (1)			4	4	3	
	(e)		Proton repulsion or like charges repel etc.	1			1		
	(f)		Photon mom calculated $\left(p = \frac{h}{\lambda}\right) = 2.73 \times 10^{-22}$ [kg m s ⁻¹] (1) Electron momentum calculated = 9.11×10^{-26} [kg m s ⁻¹] (1) [Initial momentum negligible] so final momenta must cancel (1)			3	3	2	

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	(g)		Charge of $\bar{u}d = -\frac{2}{3} - \frac{1}{3}$	1			1		
	(h)		Either: Mass = $\frac{172 \text{ G[eV]}}{931 \text{ M[eV u}^{-1}]} = 185 \text{ [u]} \text{ (1)}$ $\therefore \text{Mass} = 185 \text{ [u]} \times 1.66 \times 10^{-27} \text{ [kg u}^{-1}] = 3.07 \times 10^{-25} \text{ [kg]} \text{ (1)}$ Or: Mass energy = $172 \text{ GeV} \times 1.60 \times 10^{-19} \text{ J eV}^{-1} = 2.75 \times 10^{-8} \text{ [J]} \text{ (1)}$ $\therefore \text{Mass} = \frac{2.75 \times 10^{-8} \text{ [J]}}{(3.00 \times 10^8 \text{ [m s}^{-1}])^2} = 3.06 \times 10^{-25} \text{ [kg]} \text{ (1)}$		2		2	2	
			Question 7 total	3	8	9	20	12	0